

Osama Ali

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SUMMARY

AI/ML Engineer with hands-on experience in Machine Learning, Generative AI, and NLP. Proven track record in building and fine-tuning models for real-world applications through two successful industry internships. Focused on shipping production-ready features over prototypes. Passionate about developing intelligent, scalable systems that deliver tangible results.

EDUCATION

Bachelor of Science in Computer Science

FAST National University of Computer and Emerging Sciences (NUCES)

Aug 2021 – Dec 2025

Islamabad, Pakistan

SKILLS

ML and AI: Deep Learning, Generative AI, Large Language Models, NLP, RAG, Scikit-learn, PyTorch, LangChain, LangGraph, Fine-Tuning, CI/CD

Data Science: Python, SQL, Pandas, NumPy, Matplotlib,

Tools/Libraries: Flask, REST APIs, Git, Docker, Selenium, FastAPI, React

WORK EXPERIENCE

AI Engineer Intern

Buildables

Dec 2025 – March 2026

Islamabad, Pakistan

- Contributed to an enterprise Urdu speech-to-speech chatbot for local farmers; developed training pipeline and contributed to speech model fine-tuning. Reducing WER from 35% to 10%.
- Built a text-to-image generation service, exposed via gRPC and FastAPI endpoints handling 10 concurrent requests.
- Containerized with Docker and deployed as a public API on Hugging Face Spaces with automated CI/CD via GitHub Actions.

Machine Learning Intern

Bytewise

Jun 2024 – Aug 2024

Islamabad, Pakistan

- Designed churn prediction system using Logistic Regression, Decision Trees, and XGBoost; deployed with Flask API.
- Conducted data preprocessing, model training, and performance evaluation on datasets for classification and prediction tasks.

PROJECTS

ClearFrame: Real-Time Misinformation Detection System

[View Project](#)

Technologies Used: Python, FastAPI, React.js, PostgreSQL, PyTorch, Hugging Face Transformers, RAG, BeautifulSoup, Selenium

- Built an AI-powered real-time fact-checking system for Twitter (X) and Facebook, focused on Pakistan-relevant multilingual (Urdu–English–Roman Urdu) misinformation posts.
- Fine-tuned XLM-RoBERTa model on **20K scrapped posts to achieved accuracy of 90%**.
- Designed modular FastAPI backend with PostgreSQL, integrating NLP pipelines, Multilingual cross-verification, and crowdsourced verification.
- Integrated Hugging Face models for entity recognition and inference; implemented RAG pipelines with Google Custom Search API and custom scraping scripts for dynamic claim validation.
- Achieved average verification **response time of ~10 seconds** end-to-end.

DocChat: Multi-Agent RAG

[View Project](#)

Technologies Used: Python, LangGraph, Google Gemini (Pro Embeddings), Docling, ChromaDB, BM25, Gradio

- Developed a multi-agent RAG system for hallucination-resistant question answering over complex documents (PDF, DOCX), using a LangGraph workflow with 4 specialized agents: Relevance Checker, Research Agent, Verification Agent, and a self-correction loop.
- Implemented hybrid retrieval combining BM25 keyword search and Gemini semantic embeddings, achieving a **7% improvement in retrieval accuracy** over semantic-only search on tables, footnotes, and technical content.
- Integrated IBM's Docling parser for OCR, Markdown structuring, and header-based chunking; added ChromaDB vector store with caching to avoid reprocessing unchanged documents.

CERTIFICATIONS

- Gen AI with LLMs – IBM
- Machine Learning Specialization – DeepLearning.AI
- Deep Learning Specialization – DeepLearning.AI
- Cloud Foundations – AWS Academy

EXTRA CURRICULAR

- Organizing member of the TensorFlow Community Islamabad, coordinating tech seminars and virtual events.